MOHAMED AYMAN EL-HALMOUSHY

AI & Intelligent Systems Engineering

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Summary

Top-ranked Intelligent Systems Engineering undergraduate with a 3.79 GPA and expertise in AI, computer vision, and radar-based detection. Skilled in Python, C++, MATLAB, and embedded systems, with demonstrated success optimizing ML& DL models for real-time applications.

Work Experience

AI Training Intern (*Telecom Egypt (WE) | Cairo, Egypt***)** Jan 2024 - Feb 2024

Telecom Egypt is a leading telecommunications provider in Egypt

- Enhanced Python-based AI models for object detection, segmentation, and classification
- Trained and optimized ML/DL models (TensorFlow) for real-time applications
- Boosted object detection accuracy to 93% (from 80%) using YOLOv8 fine-tuning

Projects

1- OncoEgypt System – Cancer Patient Monitoring & Automation

- Designed and developed an automated system for monitoring cancer patients, managing treatment session schedules, and tracking doctors' evaluations of patient health status.
- Built the system using **n8n** automation tools integrated with WhatsApp to enable continuous and real time patient follow up.
- Created automated workflows for sending appointment reminders, collecting patient updates, and improving communication efficiency.
- Implemented a data management solution using Google Sheets for organized, accessible, and scalable data storage.
- Enhanced system functionality using **Google Apps Script** for automation, data processing, and integration.
- Successfully deployed the system in a real world environment for a medical researcher specializing in oncology (Master's level).

2- Aang – AI Avatar for Kids (Hackathon Project)

Developed an AI-powered avatar that teaches children about AI through interactive cartoons with real-time animation. add TTS/STT tm make model respond in chat using messages and voice.

- Secured **2nd place** among 36 universities at the Helwan Cyber Arena Hackathon, winning 30,000 EGP, organized by **Cyshield & EG-CERT**.

3- Real-Time Object Detection Using SSD Model

A project focusing on the development of an object detection system Developed a real-time object detection system using SSD model with OpenCV and TensorFlow

- **Reduced inference time by 30%**

4- Car Detection, Segmentation, and Counting in Tahrir Square

Project focusing on traffic analysis using AI Developed an AI system using YOLOv8 to detect, segment, and count cars in video footage from Tahrir Square

- Detected 200+ cars/minute with 90% accuracy; reduced false positives by 20%

5- Attack Me If You Can – AI Cyber Defense Multi Agent System

network environment using a collaborative multi agent architecture.

The system consists of four specialized AI agents working together:

- **Intrusion Detection Agent (IDS):** Detects known attacks such as DDoS, port scanning, brute force, and SQL injection using machine learning models (XGBoost).
- **Anomaly Detection Agent:** Identifies abnormal behavior and zero-day attacks using Isolation Forest.
- **Log Analysis Agent:** Analyzes system logs and extracts threat-related patterns using NLP models like DistilBERT.
- **Response Agent:** Takes automated actions such as blocking IPs, sending alerts, or monitoring suspicious activity.
- A custom **Risk Fusion Engine** combines outputs from all agents using weighted scoring to produce a final decision, improving accuracy and reducing false positives

Core Skills

- Programming Languages: Python, C++,java script,java, SQL.
- Machine Learning & AI: Supervised learning (regression, classification) , Deep learning (CNN, ANN, LSTM) , Model evaluation (F1-Score, MAE) ,Model tuning, transfer learning and optimization
- Frameworks & Libraries: TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn.
- Data Handling & Analysis: Data cleaning, preprocessing, outlier handling, Feature engineering and transformation, Real-time ML pipelines with FastAPI and Gradio, Flask API, Business Intelligence & Visualization, SQL querying, Power BI dashboard design, Excel for data analysis.
- Tools & Environments: Jupyter Notebook, Anaconda, GitHub for version control, Hugging Face & Kaggle for deployment and collaboration, n8n.
- Soft Skills: Problem-Solving, Teamwork, Time Management, leadership and mentoring, Cross-functional teamwork and peer collaboration, Technical presentation and documentation preparation.

Education

Helwan National University Sep 2022 - Feb 2027

B.Sc. Intelligent Systems Engineering

GPA: 3.79 (First Rank)

Languages

- Arabic (Native)
- English (Advanced)

Awards

2nd Place AI Helwan Cyber Arena Hackathon (AI Competition) (Cyshield & EG-CERT)

Certificates

- **Artificial Intelligence (HCIA(Score: 98%)) (NTI-Huawei Egyptian Talents Academy (ETA))**
- **Deep Learning for computer vision(Score:98%) (NTI Digital Egypt Youth)**
- **Introduction to Deep Learning & Neural Networks with Keras** (IBM & Coursera)
- **Machine Learning with Python** (IBM & Coursera)
- **Microsoft Machine Learning Engineer** (DEPI)
- **Internet of Things (IoT) Applications** (NTI)
- **Internet of Things (IoT) Applications development** (ITI)
- **How AWS Managed Services (AMS) works within cloud operating models** (AWS Training Center)
- **Design Thinking & Entrepreneurship** (Innov Egypt (ITIDA))